

Graph $G(V, E)$

$$V - E + R = 2$$

Euler circuit

T is a graph $\begin{cases} \textcircled{1} \text{ connected} \\ \textcircled{2} \text{ no cycle} \end{cases}$

Spanning subgraph
tree

Thm given a T a.b unique path

 但 def no cycle $\times$

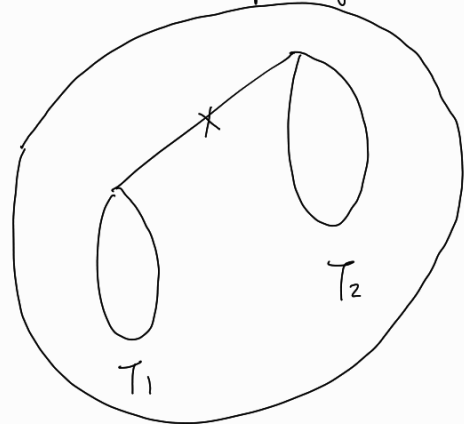
$|V| = |E| + 1$ <pf> By ^{Math} Induction

$T = (V, E)$

$$|E| < k$$

$$|E| = k$$

全部 edge = $|E|$



$$|V_1| = |E_1| + 1$$

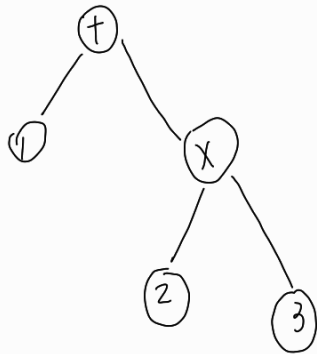
$$|V_2| = |E_2| + 1$$

$$\begin{aligned} |V| &= |V_1| + |V_2| + |E_1| + |E_2| + 1 \\ &= |E| \end{aligned}$$

infix ?

$$1+2 \times 3$$

expression tree



Arithmetic

$$1+2$$

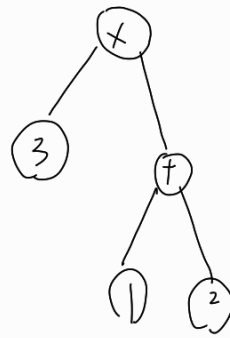
$$+ 1 2$$

prefix

$$1 2 +$$

postfix

$$(1+2) \times 3$$



$$(1+(2 \times 3))$$

$$+ 1 \times 2 3$$

$$1 2 3 \times +$$

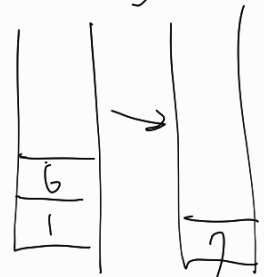
$$(1+(2 \times 3))$$

$$) \circ +$$

$$1 (2 3 \times) +$$



stack



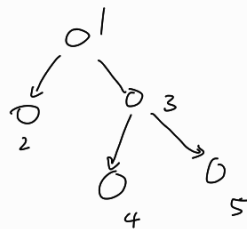
traversal | 每個node都要走得

② ① ④ ③ ⑤

inorder

preorder

postorder



lexicograph 字典排序法

Coding



目的:
 { efficient
 redundant
 security }

parity check

101010 □ { 奇数個 1 → 1
 偶 → 0

001 0.06 A 00 11
 01 0.15 B 01 10
 1 0.8 C 10 01

期程 2

A B C 000 110

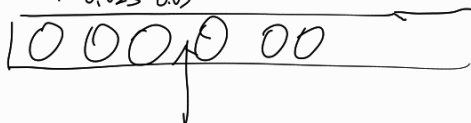
fixed-length coding

variable-length coding

Halfman coding

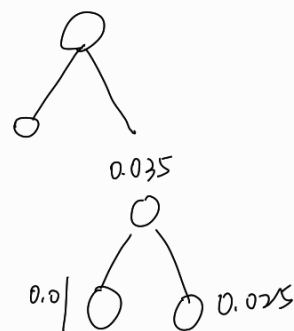
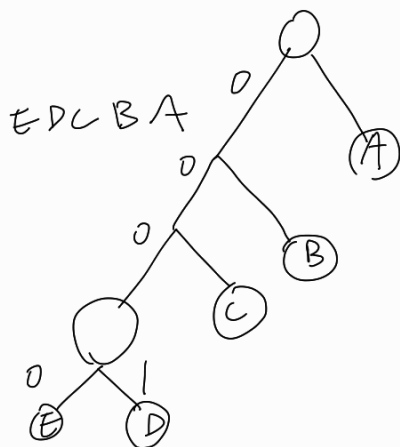
A 0.1
 B 0.4
 C 0.2
 D
 E

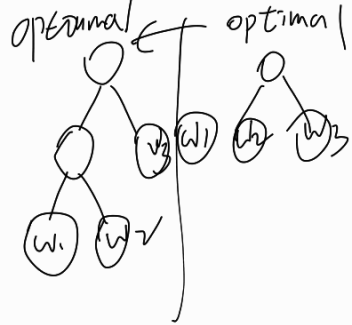
機率從小排到大
 0.01 0.025 0.03



Ex

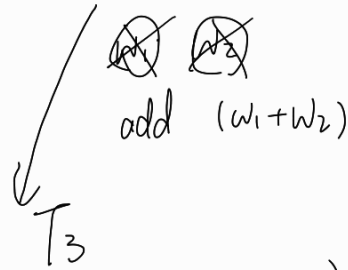
1 A 0.06
 01 B 0.02
 001 C 0.015
 0001 D 0.004
 00001 E 0.001





If $T: (w_1 + w_2) w_3 \dots$
 , then $T_1: w_1 w_2 w_3$

<pf> Let $T_2: \text{optimal} \mid w_1 w_2 w_3$



$$W(T_2) = W(T_3) + w_1 + w_2$$

$$W(T_1) = W(T) + w_1 + w_2$$

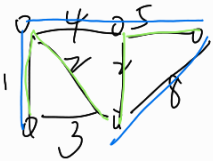
$\therefore T: \text{optimal}$

$$\therefore W(T) \leq W(T_3)$$

$$\text{if } W(T) < W(T_3)$$

$$W(T) < W(T_2) \quad \times$$

minimal Spanning Tree



$$1 + 2 + 2 + 5 = 10$$

$$1 + 4 + 5 + 8 = 18$$

Graph

Find the minimal Spanning tree

Kruskal's Alg

By edge

sorting edge by ascending & 但不形成 cycle

Prim's Alg

By vertex



Kruskal's Algmn proof

T : spanning tree by Kruskal's Algmn

$e_1 e_2 e_3 e_4 \dots e_k \dots$

optimal T' $e_1 e_2 e_3 \dots e_{k'} \dots$

$$d(T, T') = k$$

T : Kruskal

T_1 : optimal

$$d(T, T_1) = r$$

$$\textcircled{1} \quad r = n \rightarrow T_1 = T$$

$$\textcircled{2} \quad r \leq n-1 \quad T: \boxed{e_1 e_2 \dots e_{r-1}} e_r \dots$$

$$T_1: \boxed{e_1 e_2 \dots e_{r-1}} e_r \dots$$

$$T_2 \quad e_1 e_2 \dots e_{r-1} e_r \quad \cancel{e_r'}$$

$$\therefore W(T_2) = W(T_1) + W(e_r) - W(e_r')$$

$$W(e_r') \geq W(e_r)$$

$$\therefore W(T_2) \leq W(T_1)$$

$$W(T_2) = W(T_1)$$